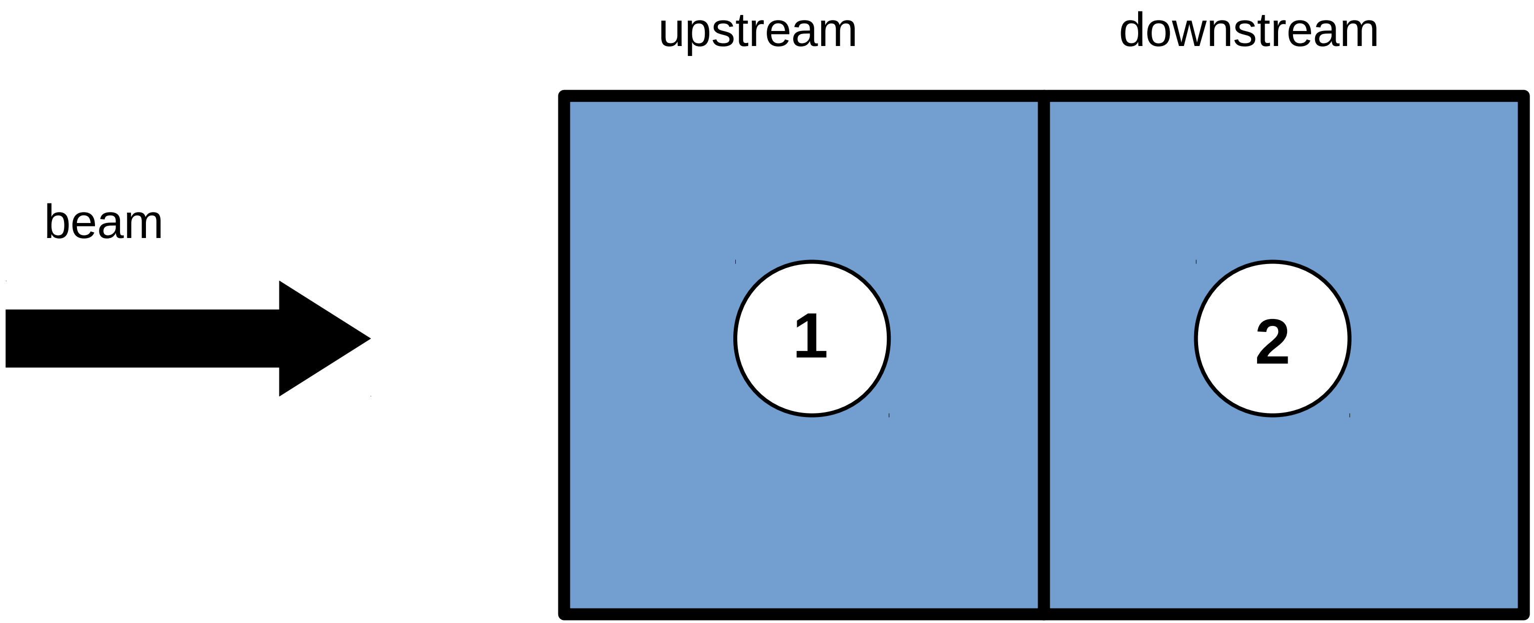
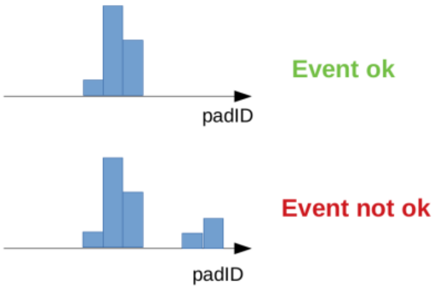
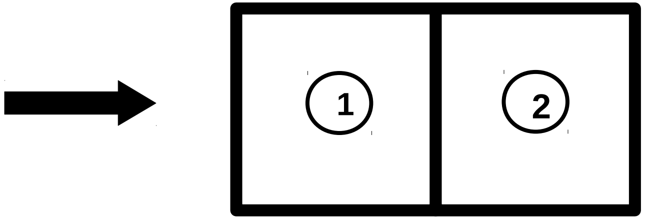


MWPC1 event selection:

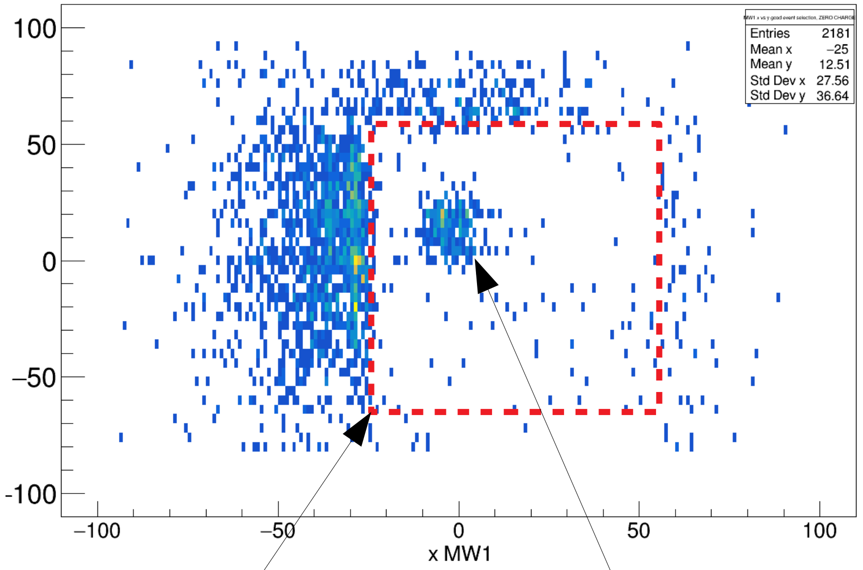
Nr. hits in x and y < 5



Case A: $Z_1 = 0$; $Z_2 = 0$



MW1 xy



Geom. acceptance

??

Ok, I made **strict** cut regions

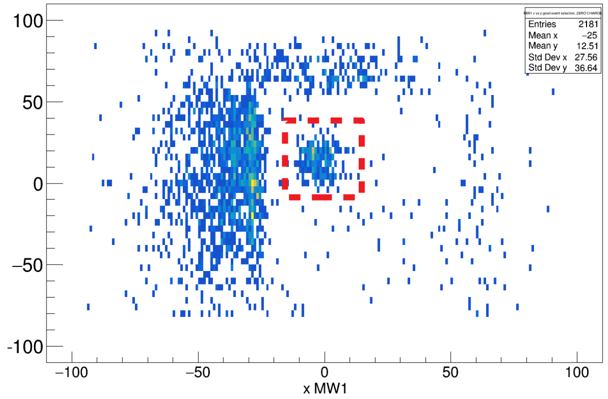
Inside TWIN:

→ **$-15 < x < 15$**

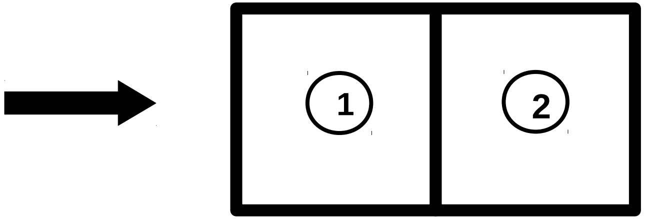
→ **$-7 < y < 46$**

Outside TWIN:

else

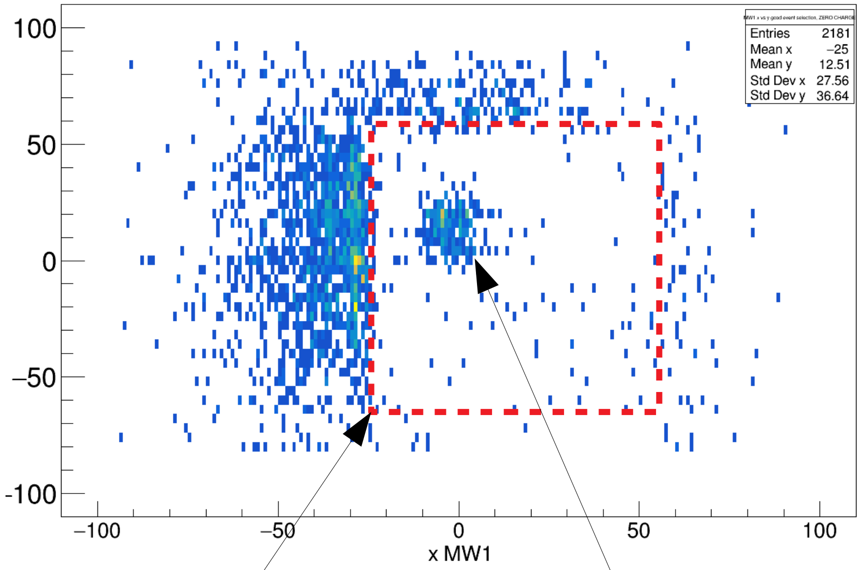


Case A: $Z_1 = 0$; $Z_2 = 0$

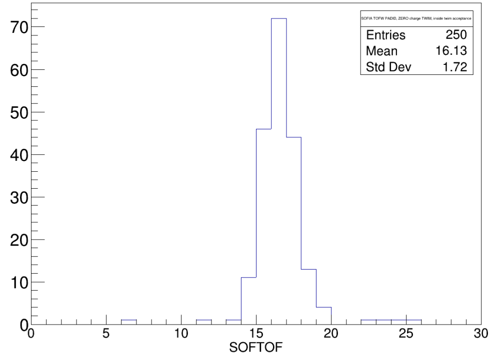


Sofia ToFW PADID

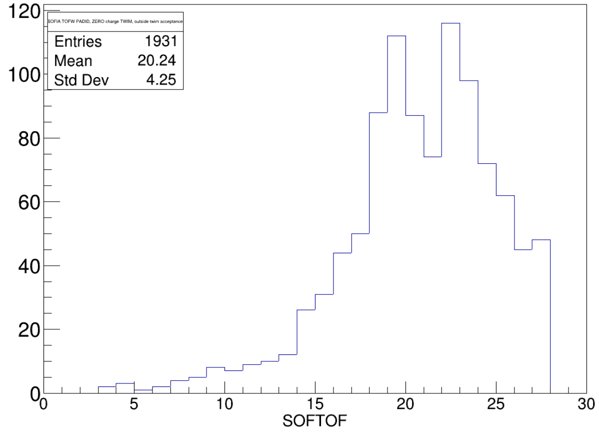
MW1 xy



Inside



outside

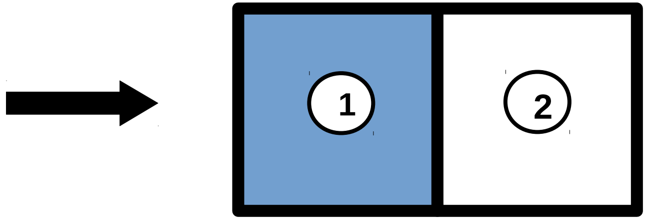


MESSEL **WIXH.**

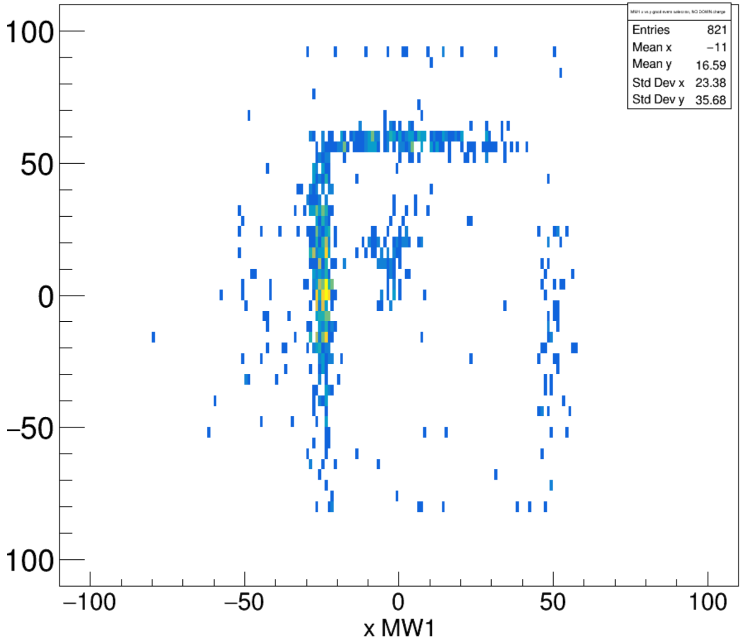
Geom. acceptance

??

Case B: $Z_1 > 0$; $Z_2 = 0$

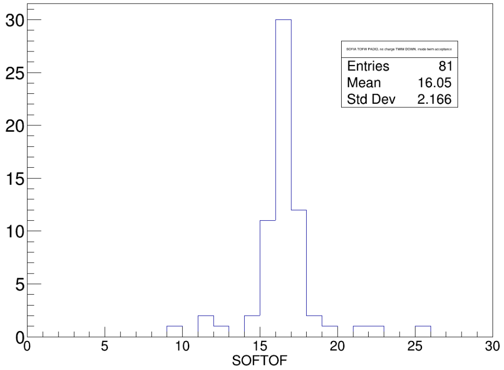


MW1 xy

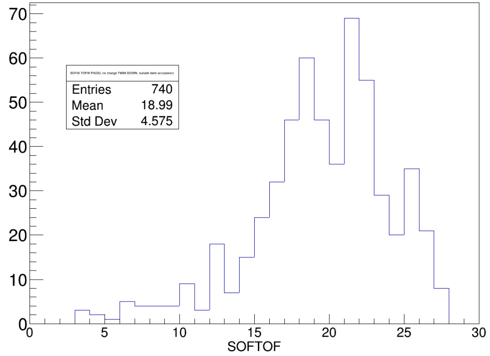


Sofia ToFW PADID

Inside

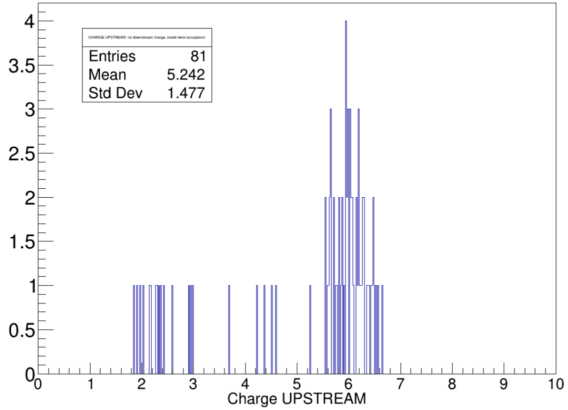


outside

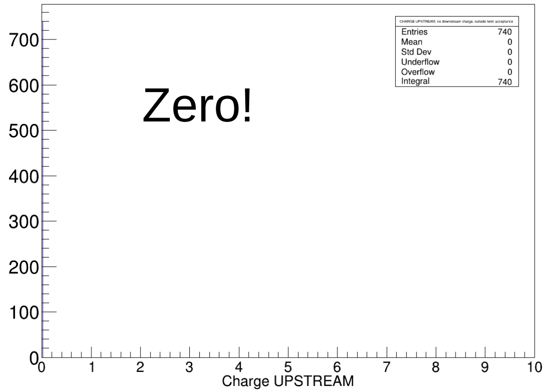


CHARGE 1), upstream

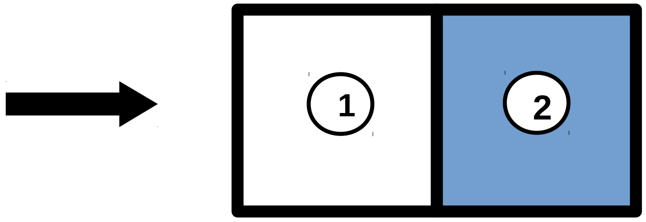
Inside



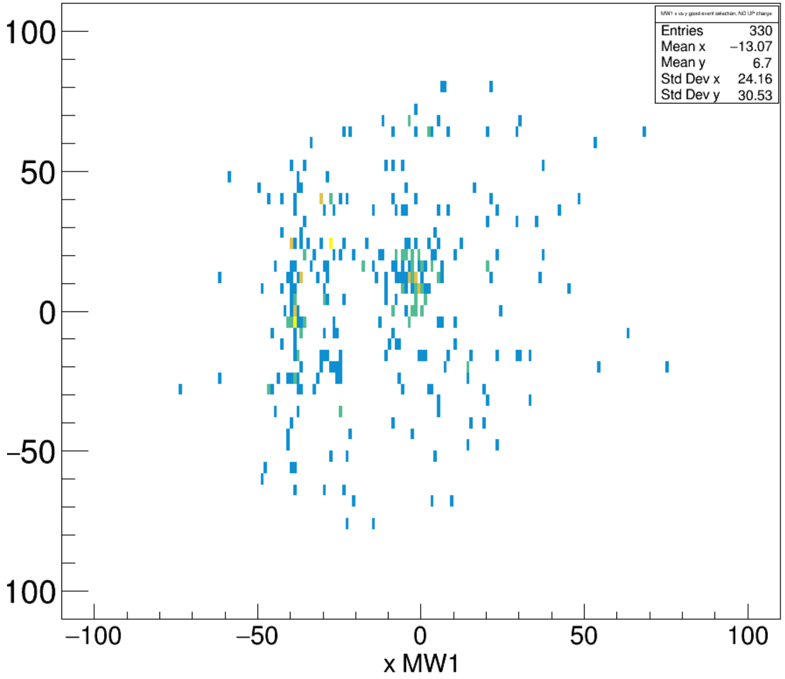
outside



Case C: $Z_1 = 0$; $Z_2 > 0$

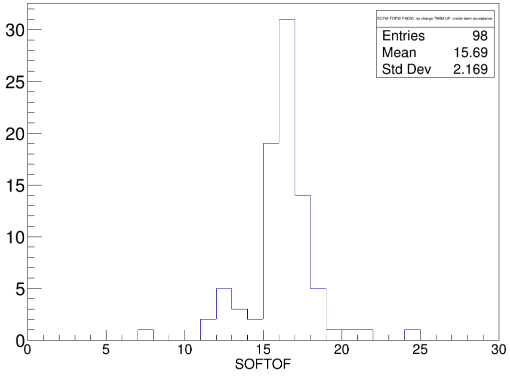


MW1 xy

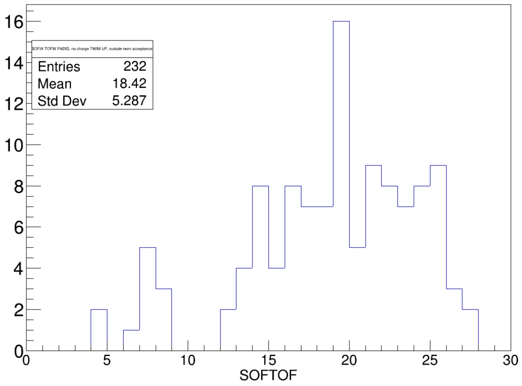


Sofia ToFW PADID

Inside

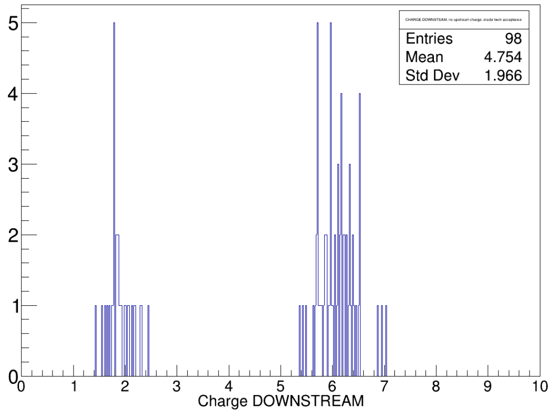


outside

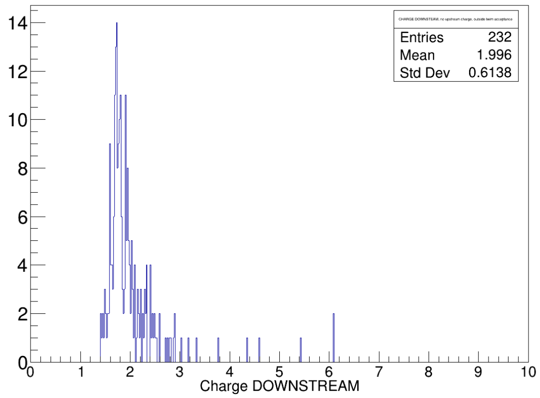


CHARGE 2), downstream

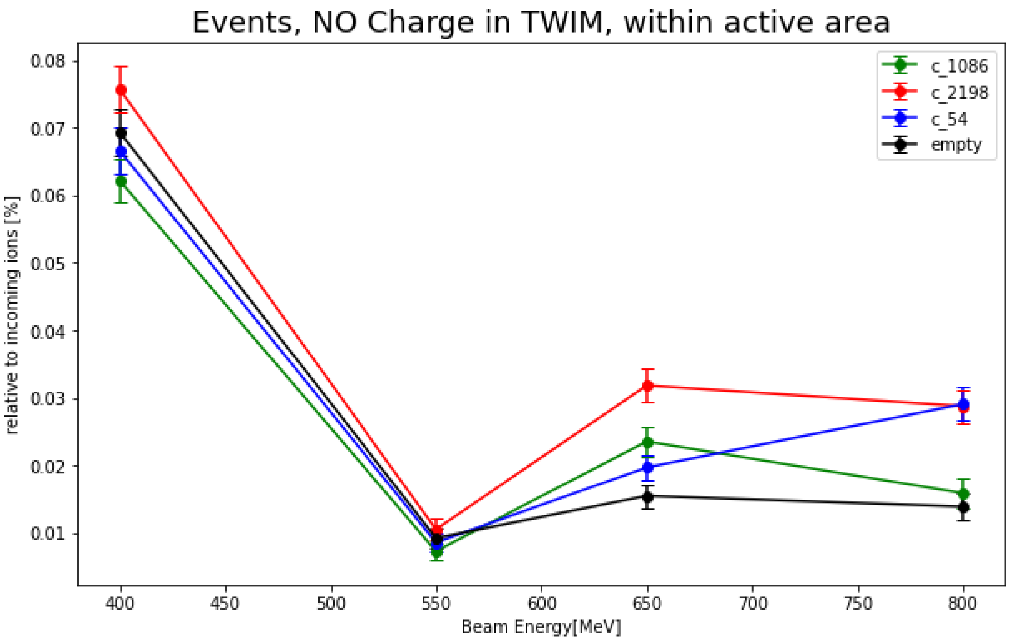
Inside



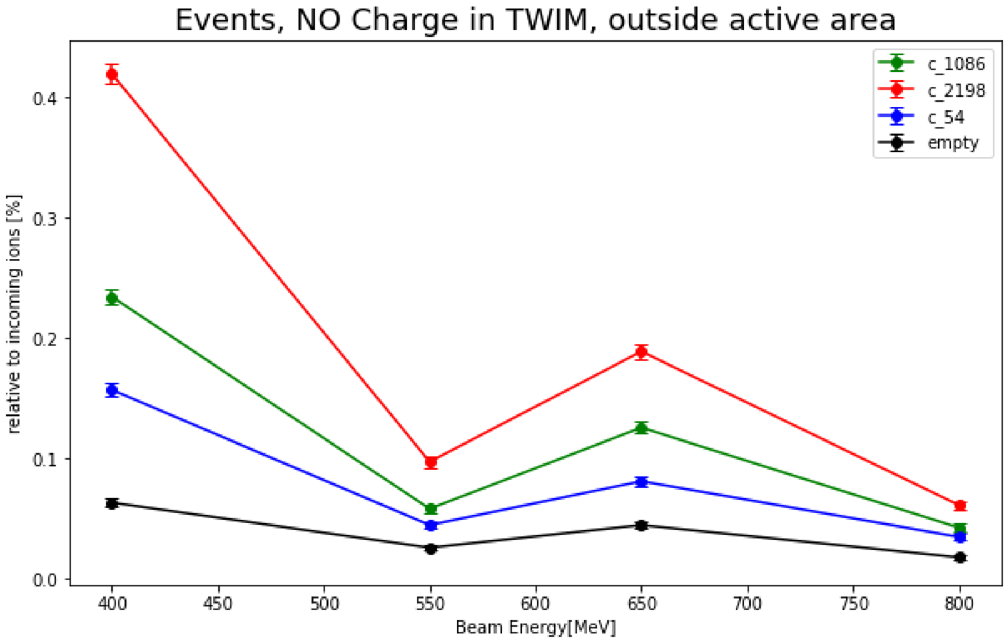
outside



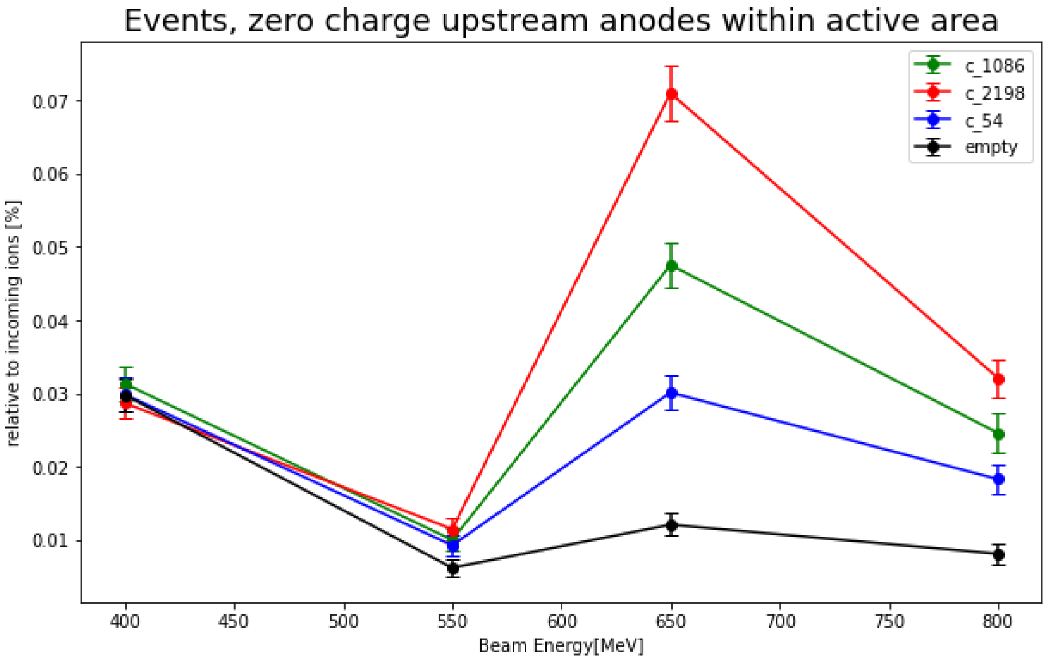
Numbers



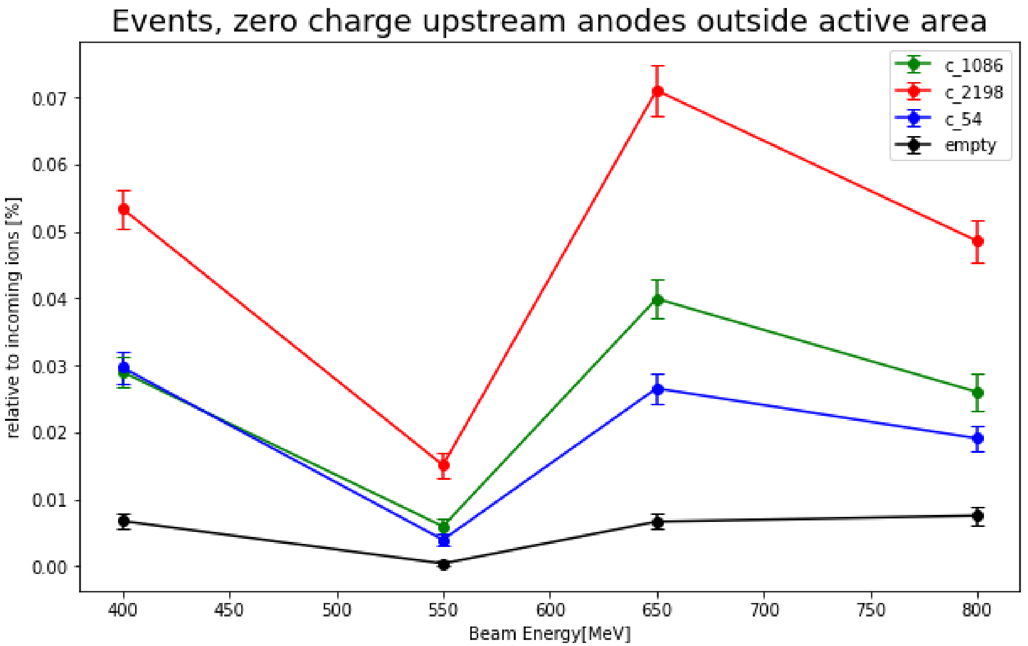
Events, NO Charge in TWIM, within active area
c_1086
c_2198
c_54
empty
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
relative to incoming ions [%]
400
450
500
550
600
650
700
750
800
Beam Energy[MeV]



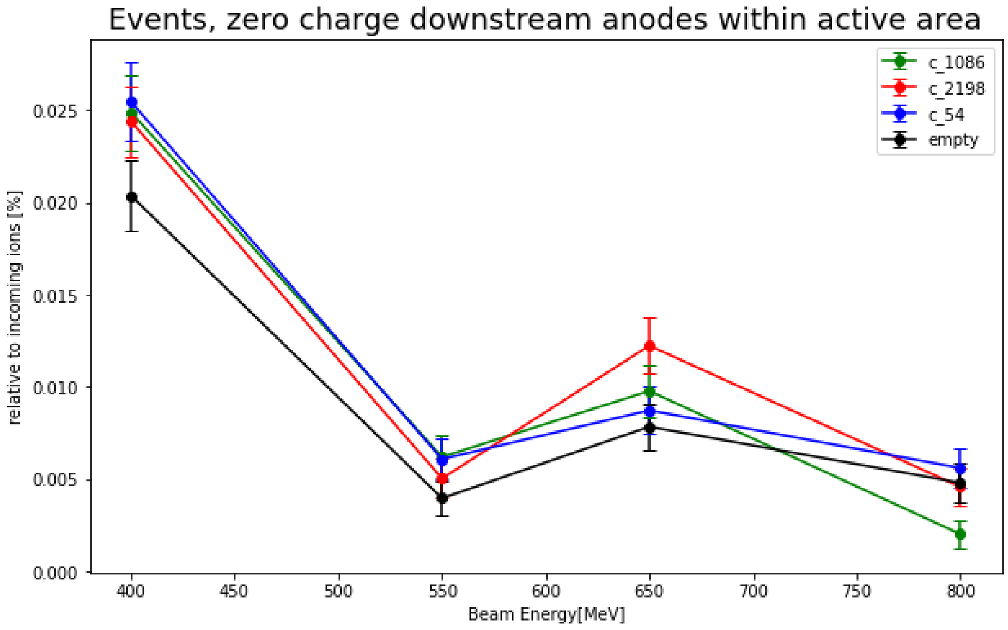
Events, NO Charge in TWIM, outside active area
c_1086
c_2198
c_54
empty
0.4
0.3
0.2
0.1
0.0
relative to incoming ions [%]
400
450
500
550
600
650
700
750
800
Beam Energy[MeV]

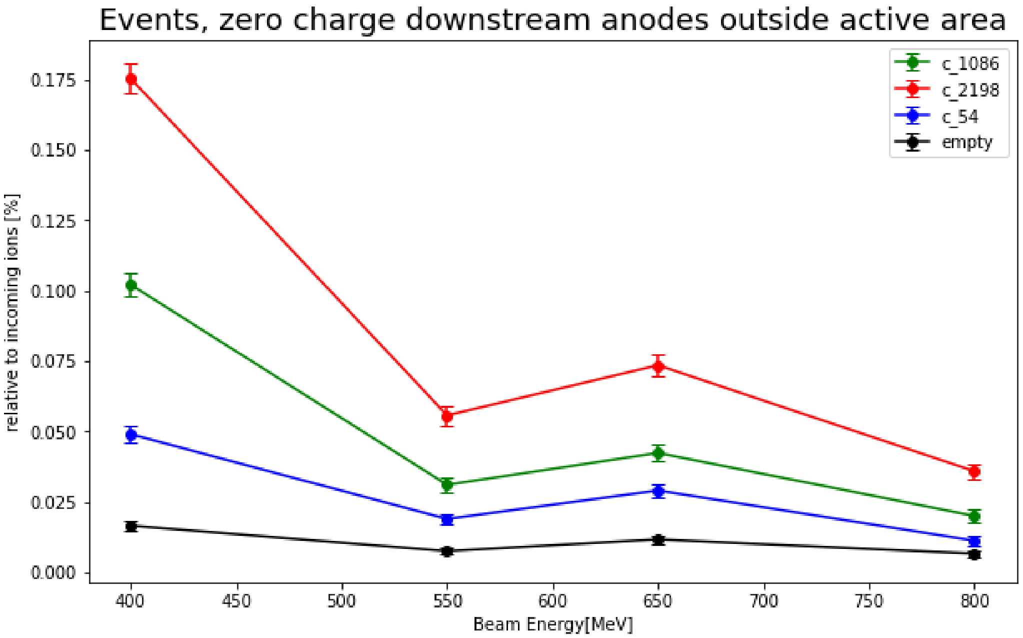


Events, zero charge upstream anodes within active area
c_1086
c_2198
c_54
empty
relative to incoming ions [%]
0.07
0.06
0.05
0.04
0.03
0.02
0.01
400
450
500
550
600
650
700
750
800
Beam Energy[MeV]



Events, zero charge upstream anodes outside active area
c_1086
c_2198
c_54
empty
relative to incoming ions [%]
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
400
450
500
550
600
650
700
750
800
Beam Energy[MeV]


Events, zero charge downstream anodes within active area
c_1086
c_2198
c_54
empty
relative to incoming ions [%]
0.000
0.005
0.010
0.015
0.020
0.025
400
450
500
550
600
650
700
750
800
Beam Energy[MeV]


Events, zero charge downstream anodes outside active area
c_1086
c_2198
c_54
empty
relative to incoming ions [%]
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
400
450
500
550
600
650
700
750
800
Beam Energy[MeV]